



The Validation of Auto Rickshaw Model for Frontal Crash Studies Using Video Capture Data

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Abstract

Despite being Auto rickshaws are the most important public transportation around Asian countries and especially in India, the safety standards and regulations have not been established as much as for the car segment. The Crash simulations have evolved to analyze the vehicle crashworthiness since crash experimentations are costly. The work intends to provide the validation for an Auto rickshaw model by comparing frontal crash simulation with a random head-on crash video. MATLAB video processing tool has been

used to process the crash video, and the impact velocity of the frontal crash is obtained. The vehicle modelled in CATIA is imported in the LS-DYNA software simulation environment to perform frontal crash simulation at the captured speed. The simulation is compared with the crash video at 5, 25, and 40 milliseconds respectively. The comparison shows that the crash pattern of simulation and real crash video are similar in detail. Thus the modelled Auto-rickshaw can be used in the future to validate the real-time crash for providing the scope of improvement in Three-wheeler safety.

Introduction

Accidents in developed countries are considerably lesser when compared to the developing countries, which is 70 times the likeness of occurrence. Besides four-wheelers, the Auto rickshaw is used in Asian countries which the majority of the people used for transportation [5]. Three-wheeler accident result is results up to 5% of reported fatalities in India. Even though the numbers are lesser, the impact is more. But, there is not much research has been carried out considering the safety of this vehicle [5]. As a first attempt, the three-wheeler vehicle crash characteristic was analyzed using crash simulation software (MADYMO) with the intent to improve the safety features with pedestrians and other road users in 1997. The attempt to improve the crashworthiness of these vehicles are not being tried before [1]. It is essential to apply finite element analysis to design and evaluate the crashworthiness of the vehicle model to improve safety. The Automotive crash analysis software like LS-DYNA and MADYMO can be a vital tool to simulate the reality of events [3]. Modern auto rickshaws are made to run at a higher speed to match the demands of the customer needs, and hence the safety is a significant concern. As the speed gets higher, the result of fatalities also creeps in numbers over the years [3]. Auto rickshaws are generally a low-speed vehicle which has a maximum speed of 60 km/hr. For crash simulation environment, and the international crashworthiness standard speed is set between 20 km/hr to 50 km/hr [3]. In real-time, the auto-rickshaw crash occurs

either by impacting a static or dynamic obstacle, which collides with the vehicle of the same size, even bigger or size of the other object in the environment [3]. Auto rickshaw at velocities of 20 km/hr or even more causes injuries which are of high risk to the road users. Side offset impact produces lower risk scenario in which no head contact is recovered in pedestrian auto-rickshaw crash simulation [4]. Despite decades of crash analysis research, frontal collisions are the reasons for approximately half of all the fatal accident [10]. The motion detection of the vehicle during a collision event helps to understand the interaction of the vehicle during crashes in an even better way. If motion detection is set up in a laboratory environment, accelerate meters and gyroscope are fixed inside the vehicle to get the accidental data. In reality, the crashes are recorded in CCTV footage and other video references. Multiple techniques are emerged to process any frame rate videos. Among other methods, MATLAB- SIMULINK image and video processing using cameras can be utilized to analyze the accident videos [7]. The motion of the moving vehicle also can be obtained by drawing reference lines forming the frame image obtained from the side-view camera [8]. Vehicle detection and vehicle tracking are vital steps in speed estimation [9]. The primary idea is to provide a vehicle model which resembles the vehicle involved in a real crash. The motive for the present study is to validate a modelled auto-rickshaw for crash safety analysis. MATLAB interface and LS-DYNA simulation environment are used to compare the real crash and simulated crash.

Methodology

The Overview of the study is as shown in Figure 1. The CATIA generated model is imported in the LS-DYNA crash simulation environment, and input parameters such as meshing, constraints, thickness, and contacts were given.

On the other hand, a video reference from the internet is processed in MATLAB to obtain the velocity of the Auto rickshaw. The Obtained velocity is given as the input parameter for crash simulation. If the crash pattern of the simulation and video reference is similar, then the model is validated. If not, the input parameters are needed to be changed to validate the model.

Measurement and Modelling

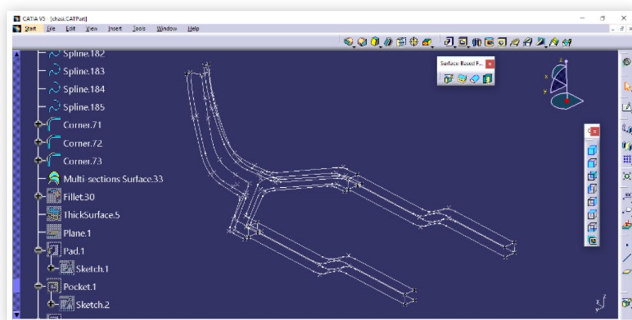
The Overall Dimensions of 'BAJAJ RE Compact' Auto rickshaw were physically measured using thread, measuring tape, and scale. The measurements were marked in the form of a scalar diagram, with that reference, the CATIA wireframe model of the Frontal body, Back body, Chassis, wheel joints, Seat, and the tire was formed. The wireframe is converted into surface and solid generation to develop the whole structure of the auto-rickshaw.

Various stages of Modelling is done in the CATIA software:

1. Wire Frame Generation
2. Surface Generation
3. Solid Generation

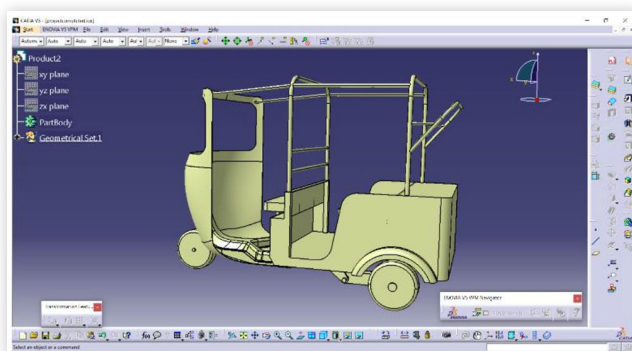
As the measurements were taken noted, extremities of points, bends, and curvature is marked. After that, the corresponding lines are formed to get the overall structure. For example, the wireframe model of Chassis is as shown in Figure 2, which displays the points and lines are constructed. Wireframe structure is the initial stage of modelling in which the basic profile is formed for each part. After that, the surface

FIGURE 2 Wireframe model of Auto rickshaw chassis



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FIGURE 3 Finite Element model of Auto rickshaw



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is generated, and the corresponding thickness was given. The parts were named separately and assembled in CATIA; the whole assembly of an auto-rickshaw is as shown in figure 3. The list of parts assembled are listed below

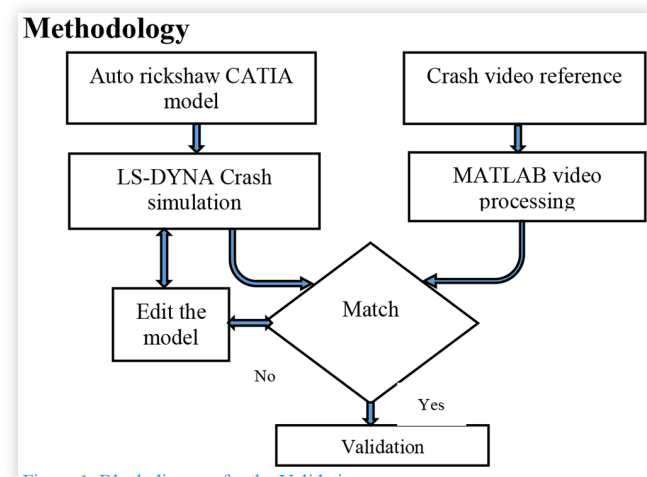
1. Front tire
2. Wheel connector
3. Frontal panel
4. Chassis
5. Windshield frame
6. Back body
7. Seats
8. Body frame
9. Rear joints

Frontal mudguard affects the impact of dynamics during the frontal crash. The auto-rickshaw model without a mudguard induces significant shearing of the frontal body [6]. Therefore, the mudguard is neglected from the whole model.

MATLAB Video Processing

A video from the internet is taken as a reference. MATLAB interfaces make provision for object detection and video processing method which is the new aspect of understanding the crash dynamics in a better way. The idea is to find a crash video from the internet and processing them to see the

FIGURE 1 Block diagram for the Validation process



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necessary input for the crash simulation. In the video reference, the auto-rickshaw is impacting the side of the bus, and the frontal portion is completely deformed. This video resembles the head-on crash of the vehicle with the wall, as shown in [Figure 5](#). For better deformation behavior, the bus side panel is assumed as a rigid body. To find the Velocity, the total number of frames and the frame rate should be obtained first. With the frame per second and total numbers of frames, the total time in seconds is obtained using code generation. The Basic step is to find the velocity of any moving object is to convert the video into the .mp4 format and the video is uploaded in the generated code. Assuming that the starting point of Auto rickshaw is the start of the video, distance covered by the vehicle is obtained by measuring the total number of bus parking plots and width of each bus parking plot from the video approximately. The initial distance between the auto-rickshaw and the bus is expected to be 30 meters.

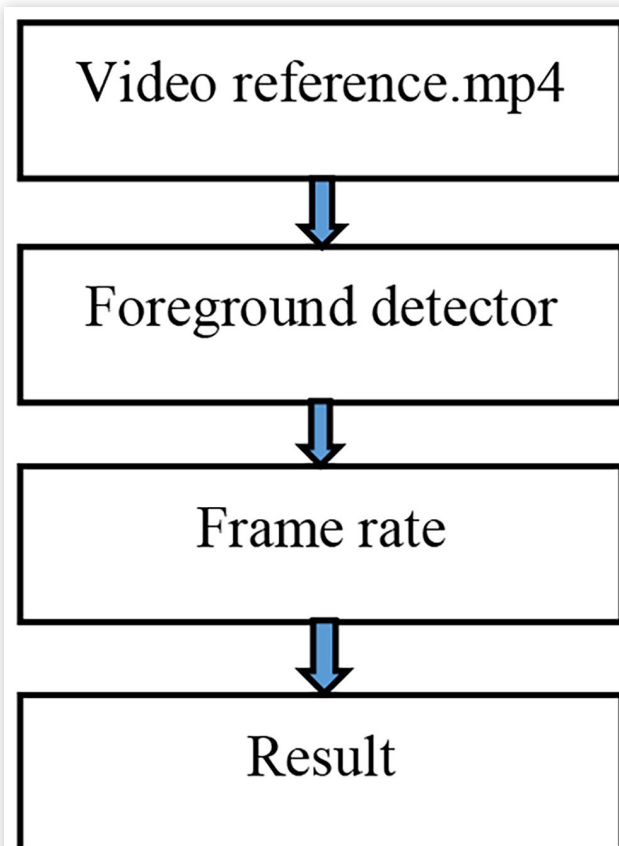
The obtained number of frames, frame rate and Distance covered by the Auto rickshaw are processed in the below formulae.

$$\text{Speed} = \text{Distance} / \text{Time}$$

$$\text{Time} = \text{Number of frames} / \text{Frame rate}$$

The structure of video processing is as shown in [figure 4](#). The velocity of moving auto rickshaw is found to be 40kmph from the video. This velocity value is given as an input for the crash simulation.

FIGURE 4 Block diagram for Velocity Estimation



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FIGURE 5 Crash video reference



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Crash Simulation

Meshing

The CATIA model is imported in LS-DYNA, and each part is meshed at different element size, considering the factors such as geometry and shape complexity as listed in [Table 1](#). The Wall and road are created in SHAPE_MESHER at coarse rectangular mesh.

TABLE 1 Mesh element size for different parts

Parts	Element Size (mm)
Frontal Body	9
Wheels	47.87
Seat	47.87
Back Body	47.87
Chassis	15
Rear Joints	30
Frame	15

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Material Definition

The materials were assigned to different vehicle parts in the LS-DYNA material database library. The keyword used for tire, wall, and road, vehicle structure is MAT_ELASTIC, MAT_RIGID, and MAT_PIECEWISE_LINEAR_PLASTICITY, respectively, as listed in [Table 2](#). The shell thickness of the parts is listed in [Table 3](#).

Thickness of each shell, metal parts are measured manually and values are entered in LS-DYNA.

Constraints

The constraint between two parts was given by the keyword NODAL_RIGID_BODY in Entity creation. This type of constraint connects a single node of one part to multiple nodes of another part, as shown in [Figure 6](#) as green lines.

Contact Definition

The contact between two parts are given to prevent penetration during the crash. There are a lot of contact types are available in the keyword manager. In our case, the contact is directly in between two surfaces. Therefore, the keyword

TABLE 2 Material properties for vehicle part set

S.No	Vehicle part set	Mass Density	Young's Modulus	Poisson's Ratio	Yield Stress
units		Kg/mm ³	kN/mm ²		kN/mm ²
1.	Vehicle Structure	7.80e-6	207	0.3	2.0
2.	Tire	1.700e-6	24.61	0.32	-
3.	Wall and Road	7.860e-12	2e5	0.3	-

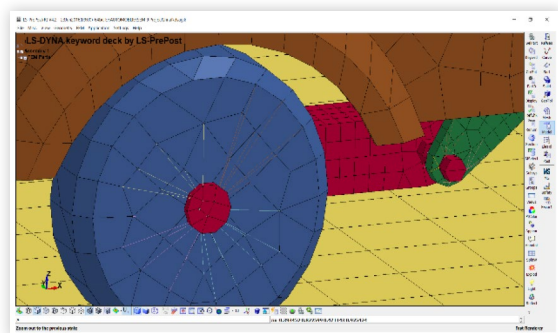
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TABLE 3 Shell thickness

S.No	Vehicle Items	Thickness (mm)
1.	Vehicle Structure	3
2.	Seat	2
3.	Wall	2
4.	Frame	2

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FIGURE 6 Nodal rigid body constraint



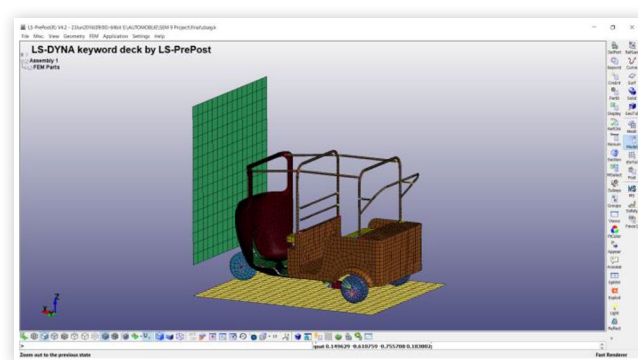
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TABLE 4 Master and slave part details

Contact id	Master part	Slave part
1	Wall	Front Tyre
2	Wall	Front Body
3	Road	Front Tyre
4	Road	Rear Tyre
5	Front Body	Chassis
6	Front Tyre	Chassis
7	Chassis	Back Body
8	Front Body	Frame
9	Back Body	Frame

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FIGURE 7 Crash simulation setup



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CONTACT_AUTOMATIC_SURFACE_TO_SURFACE is used. The master and slave parts are listed in [Table 4](#).

The keyword LOAD_DENSITY_DEPTH assigns the gravitational acceleration of the auto-rickshaw. The obtained velocity of 40 Km/hr from MATLAB video processing is given as the input parameter in the keyword VELOCITY_GENERATION. The termination time is given as 40 ms in the keyword CONTROL_TERMINATION as the vehicle comes to rest at the above mentioned time after impact in the video reference. After providing the input parameters, the simulation is initiated, and the setup of Auto rickshaw with wall and road is as shown in [Figure 7](#).

Result and Discussion

Crash Initiation Stage

The video reference and crash simulation are observed at different time stages. In [figure 8](#), in the real-time crash video, the vehicle is about to collide with the bus after 5 ms and it depicts the frontal collision with the rigid wall, and thereby the frontal simulation was carried out in LS-DYNA. At this stage, the front tire is about to intrude into the ground clearance of the bus.

Mid-Crash Stage

At 25 ms, the front tire intruded into the ground clearance as shown in [figure 9](#). The frontal connector was broken thereby

FIGURE 8 Comparison after 5 ms termination time**FIGURE 9** Comparison after 25 ms termination time.**FIGURE 10** Comparison after 40 ms termination time

it initiating the deformation in the frontal body. The windshield frame deformed like a concave bend, and thus it pulls the frame.

After the Crash Stage

At 40 ms, the front tire lost its road contact, and the frontal body is crumbled which enters into the driver's cabin. The floor of the vehicle is moved upwards since half of the Chassis also crushed. The Body frame pulls forward, which disturbed the body structure. The chassis deformation drags rear joints. There is no physical deformation observed in the back body from the simulation. The details are shown in [figure 10](#).

Summary

This study represents the methodology in which the analogy of a crash video and crash simulation is presented. The reason

behind the process is to provide an Auto rickshaw model for crash safety which can be a useful model to perform a simulation to show a different type of crashes. For validation, the idea is to compare the simulation with the real case scenario. Indeed, the closest to real cash scenario is a real-time crash video, so that the video is processed in MATLAB numerical environment. On the other side, the actual Auto rickshaw measurements were obtained to build a similar Auto rickshaw model in CATIA. The model is then imported into LS- DYNA crash simulation and input parameters like obtained velocity constraints, mesh element size, contacts, thickness, gravity, etc. inputs were entered. As mentioned in Methodology, the input parameters were modified to obtain a similar crash pattern of video and simulation.

Three stages of the crash, namely assess the final comparison; Crash initiation, mid- crash and after- crash. The deformation pattern of the frontal connector, frontal body, windshield frame, chassis, body frame, rear joints, and rear body, were observed. In the end, the comparison indicates the progressive deformation occurred as similar to that of the crash video. Therefore, modelled auto-rickshaw can be a validated model to evaluate crash safety for future work.

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Abbreviations

ms - Millisecond

mm - Millimeter

km/hr - kilometer per hour

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